

Emmanuelle Coutu-Nadeau, B.Sc., B.Eng.

emmanuelle@coutunadeau.com • www.emmanuellecoutunadeau.com
linkedin.com/in/emmanuelle-coutu-nadeau • github.com/emmacona

Education, Awards, & Scholarships

Master of Sciences, Psychology – <i>Université de Montréal</i> (GPA: 4.2 / 4.3) Neuro-AI UNIQUE Excellence Scholarship (10,000 \$)	Sep 2024 – (expected Aug 2026)
Bachelor of Software Engineering – <i>McGill University</i> (GPA: 3.71 / 4) - MES Bourses de persévérance TI (3,000 \$) - Airbus Scholarship in Engineering (1,500 \$)	Sep 2018 – May 2022
Bachelor of Science, Psychology (Honours) – <i>McGill University</i> (GPA: 3.78 / 4) First-class Honours	Sep 2015 – May 2018

Clinical Research Experience

Master Student Researcher , Neurosciences of Early Development Lab (S Lippé, PhD) – <i>CHU Ste-Justine (CRHSJ)</i> Master's Thesis: Behavioral Clustering and EEG-based Prediction of Neurodevelopmental Profiles in ASD/ADHD Using Machine Learning. - Conducted EEG recordings with children with autism and other neurodevelopmental conditions. - Administered standardized cognitive assessments with pediatric oncology patients and premature infants at follow-up (18-24 months).	Sep 2024 – Present
Research Assistant , Co-venture Lab (P Conrod, PhD) – <i>CHU Ste-Justine (CRHSJ)</i> Assisted with data entry for a study on substance use onset and adolescent cognitive development.	Nov 2017 – Jul 2018
Honours Trainee , Marc O. Martel's Pain Lab – <i>McGill University (MUHC)</i> Honours Thesis: Association between perceived injustice and prescription opioid misuse. - Conducted structured clinical interviews, data collection, and saliva sampling.	Oct 2016 – May 2019

Work Experience

Product Owner & Project Manager – <i>CareSimple</i> - Coordinated clinical and technical teams for a liver transplant research project, implementing RPM for transplant patients. - Translated client needs into technical specs for a digital health platform in the Swedish market. - Managed compliance documentation for Canadian and European health regulations. - Prototyped KPI reporting and analytics tools (Python, SQL) to support data-driven decisions.	Oct 2022 – Sep 2025
Software Developer – <i>CareSimple</i> Developed features in a web-based portal for clinicians to perform RPM (React, TypeScript).	May 2020 – Oct 2022
Data Science Intern – <i>CareSimple</i> Developed a preprocessing pipeline to standardize patient health data for artificial intelligence algorithms.	May - Aug 2019

Community Engagement

Content Manager, Student Affairs Committee – <i>UNIQUE</i> Organized events and curating research highlights to support students at the intersection of Neuroscience and AI across Quebec.	Oct 2025 – Present
Event Ambassador – <i>Fondation du MAC</i> Promoted for <i>Printemps du MAC</i>, an initiative supporting the museum through fundraising events.	Jan 2025 – Present
Software Development & Business Strategy Mentor – <i>Technovation</i> Mentored girls (ages 8–18) in mobile app prototyping and business plan development to address UN Sustainable Development Goals.	Jan 2024 – Present
Volunteer – <i>Santropol Roulant</i> Supported community food security through meal preparation, home delivery for homebound individuals, and urban agriculture initiatives.	Nov 2016 – Jun 2022
Presenter, Women in Engineering (POWE) – <i>Montreal Neurological Institute (MNI), McGill University</i> Led MNI lab visits introducing high-school girls to neuroscience and biomedical engineering.	Feb 2019

Clinical Assessment Training

Wechsler Intelligence Scale for Children (WISC-V)	Apr 2025
Wechsler Adult Intelligence Scale (WAIS-IV)	Apr 2025
Bear-Dragon Inhibition Test	Mar 2025
Identifying People Vulnerable to Suicide and Implementing Protective Measures <i>Association québécoise de prévention du suicide</i>	Sep 2024
Bayley Scales of Infant and Toddler Development-IV (Bayley-IV)	Aug 2024
Structured Clinical Interview for DSM-5 (SCID-5) – Substance withdrawal	Aug 2017

Skills

- **Languages:** Fluent in English and French (spoken and written).
- **Programming & Data Analysis:** Python, Java, C, MATLAB, React, TypeScript, R, SQL, Power BI, Excel, AI/ML.
- **Certifications:** PSPO I, HIPAA, GDPR, Cybersecurity training, Leading Change in Health Informatics (John Hopkins University, Coursera).
- **Testing:** EEG (Resting-State and stimulation tasks), Saliva Sampling.

Publications

Peer-reviewed journal article

Coutu-Nadeau, E., Arutiunian, V., Martin, C.-O., Fontaine, V., Bélanger, A.-M., Jizi, K., Knoth, I. S., Hosein, A., Davoudi, S., Elsabbagh, M., Forgeot D'Arc, B., Jacquemont, S., Lippé, S. (2026). Behavioral clustering and resting-state EEG signatures of neurodevelopmental profiles using machine learning [Manuscript in preparation]. Department of Psychology, Université de Montréal.

Arutiunian, V., Davoudi, S., Knoth, I. S., **Coutu-Nadeau, E.**, Desjardins, J., O'Reilly, C., Hosein, A., Martin, C.-O., Fontaine, V., Bélanger, A.-M., Hébert-Lalonde, N., Michaud, J., Rouleau, G., Forgeot D'Arc, B., Elsabbagh, M., Jacquemont, S., Lippé, S. (2026). Distinct electrophysiologically-defined language subtypes in neurodevelopmental disorders and their molecular, genetic, neural, and behavioral profiles [Manuscript submitted for publication]. Department of Psychology, Université de Montréal.

Verner M, Sirois A, **Coutu-Nadeau E**, Fournier C, Carriere J, Martel MO. The association between perceived injustice and opioid craving in patients with chronic pain. *J Addict Med*. 2022 Jul 21.

Published abstracts and conference presentations

Coutu-Nadeau E*, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, Elsabbagh M, Forgeot D'Arc B, Jacquemont S, Lippé S. Behavioral Clustering and Resting-state EEG signatures of Neurodevelopmental Profiles Using Machine Learning. 93e Congrès de l'Acfas (TACC Trainee Research Day on Autism and Neurodivergence); 2026 May 12; Trois-Rivières, QC, Canada.

Coutu-Nadeau E*, Arutiunian V, Martin CO, Fontaine V, Bélanger AM, Jizi K, Knoth IS, Hosein A, Davoudi S, O'Reilly C, Rouleau G, Ernst C, Lucas-Tardif C, Scorah J, Couture M, Abadie P, Joobor R, Forgeot d'Arc B, Jacquemont S, Elsabbagh M, Lippé S. Resting-state EEG signatures of ASD/ADHD: toward identifying behavioral subgroups. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, Pomerleau R, **Coutu-Nadeau E**, Knoth IS, Lippé S. Hormonal modulation of brain dynamics in menstruating adolescents and young adults: EEG insights across the menstrual cycle. Society for Psychophysiological Research Annual Meeting; 2025 Oct; Montreal, QC, Canada.

Kennedy G, Baran-Goldwax M, **Coutu-Nadeau E**, Pomerleau R, Knoth IS, Lippé S. Feeling everything: estrogen, cortisol, and sensory brain dynamics across the menstrual cycle. International Society of Psychoneuroendocrinology Annual Conference; 2025 Sep; New Orleans, LA, USA.

Herrero Babiloni A*, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. *Sleep Med*. 2019 Dec;64(Suppl 1):S152–S153.

Frimerman L*, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. *J Pain*. 2019 Mar;20(4 Suppl):S55–S56.

Herrero Babiloni A*, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and opioid craving among patients with chronic pain and prescribed opioid therapy: explorations of potential mediators. 11th Congress of the European Pain Federation; 2019 Sep; Valencia, Spain.

Herrero Babiloni A*, Frimerman L, Fournier C, **Coutu-Nadeau E**, Ross A, Lavigne GJ, Martel MO. Sleep quality and prescription opioid misuse: the potential mediating role of pain intensity, negative affect, and opioid craving. World Sleep Congress; 2019 Sep; Vancouver, BC, Canada.

Frimerman L*, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 23rd Annual McGill Pain Day; 2019 Jan; Montreal, QC, Canada.

Frimerman L*, Fournier C, **Coutu-Nadeau E**, Ross A, Martel MO. Determinants of prescription opioid craving and opioid-related hedonic feelings among patients with chronic pain prescribed opioid therapy. 38th Annual Scientific Meeting of the American Pain Society; 2019 Apr; Milwaukee, WI, USA.

Ouellet N*, Fournier C, Frimerman L, **Coutu-Nadeau E**, Martel MO. Factors associated with opioid craving among patients with chronic pain prescribed long-term opioid therapy. 14th Annual Research Day of McGill's Faculty of Dentistry; 2019 Apr; Montreal, QC, Canada.